INCH-POUND

MIL-PRF-39022/7D
25 May 2001
SUPERSEDING
MIL-PRF-39022/7C
19 July 1984

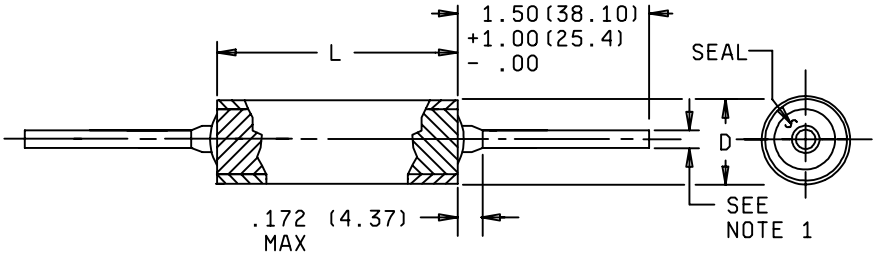
PERFORMANCE SPECIFICATION SHEET

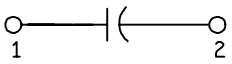
CAPACITORS, FIXED, PAPER, DIELECTRIC (METALLIZED PLASTIC FILM), ALTERNATING CURRENT (RMS) (HERMETICALLY SEALED IN METAL CASES) ESTABLISHED RELIABILITY, STYLE CHR49 (UNINSULATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39022.

INACTIVE FOR NEW DESIGN
AFTER 25 MAY 2001





CIRCUIT DIAGRAM

NOT TO BE USED WITH INSULATING SLEEVES

NOTES:
1. Number 20 AWG wire .032 ± .002 inch (0.81 ± 0.05 mm) for case diameter of .312 inch (7.92 mm). Number 18 AWG wire .040 ± .002 inch (1.02 ± 0.05 mm) for case diameter of .400 inch (10.16 mm) and over.
2. See table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only.
5. Metric equivalents are in parentheses.
6. Lead length may be a minimum of one inch (25.4 mm) long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case material: Nonmagnetic.

Capacitance value: See table I.

Capacitance tolerance: 10 percent.

Rated voltage: RMS sine wave ac at 60 through 400 Hz (see table I).

Dielectric material: Normally polycarbonate.

Operating temperature range: -55°C to +85°C.

Failure rate level: M (1.0 percent), P (0.1 percent), R (0.01 percent), and S (0.001 percent).

Thermal shock: Method 107 of MIL-STD-202, condition A .

Seal:

For unimpregnated units: Method 112 of MIL-STD-202, condition A, except at +85°C +4°C, -0°C.

For impregnated units: One hour at rated temperature with no evidence of leakage.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202.

Terminal to terminal: 140 percent of rated ac voltage (V rms at 60 Hz) on voltages of ≤ 300 volts; 120 percent of rated ac voltage (V rms at 60 Hz) on voltages of ≥ 350 volts.

Terminals to case: 900 V rms (at 60 Hz).

Insulation resistance (IR): Method 302 of MIL-STD-202, condition A.

Terminal to terminal: See table III.

Terminal to case: 10,000 megohms, minimum.

Capacitance: Method 305 of MIL-STD-202.

Dissipation factor (DF):

At 25°C: 0.3 percent maximum.

At 85°C: 0.6 percent maximum.

Barometric pressure (reduced): Method 105 of MIL-STD-202.

120 percent of rated ac voltage (400 Hz) at .82 inch of mercury (80,000 feet). See MIL-PRF-39022 for voltage limitations.

Vibration, high frequency: Method 204 of MIL-STD-202, condition B (15 G).

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Salt solution: 5 percent.

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TABLE I. Electrical characteristics, dimensions, and dash numbers.

AC rated voltage (Volts (rms))	Capacitance value (µF)	Dimension, nominal 1/ (inches)		Dash number Failure rate level for (%/1,000 hr)			
		D ± 0.062	L + 0.031 - 0.051	M (1.0)	P (0.1)	R (0.01)	S (0.001)
400	.010	0.312	1.125	1171	1191	1211	1231
400	.012	0.400	1.125	1270	1290	1310	1330
400	.015	0.400	1.125	1172	1192	1212	1232
400	.018	0.400	1.125	1271	1291	1311	1331
400	.022	0.400	1.125	1173	1193	1213	1233
400	.027	0.400	1.375	1272	1292	1312	1332
400	.033	0.400	1.375	1174	1194	1214	1234
400	.039	0.562	1.125	1273	1293	1313	1333
400	.047	0.562	1.125	1175	1195	1215	1235
400	.056	0.562	1.375	1274	1294	1314	1334
400	.068	0.562	1.375	1176	1196	1216	1236
400	.082	0.562	1.625	1275	1295	1315	1336
400	.10	0.562	1.625	1177	1197	1217	1237
390	.12	0.670	1.625	1178	1198	1218	1238
390	.15	0.670	1.625	1179	1199	1219	1239
385	.18	0.670	1.875	1276	1296	1316	1336
385	.22	0.670	1.875	1180	1200	1220	1240
380	.27	0.750	2.375	1277	1297	1317	1337
380	.33	0.750	2.375	1183	1203	1223	1243
370	.39	1.000	1.875	1278	1298	1318	1338
370	.47	1.000	1.875	1181	1201	1221	1241
350	.56	1.000	2.375	1279	1299	1319	1339
350	.68	1.000	2.375	1182	1202	1222	1242
300	0.10	0.500	1.125	1044	1074	1104	1134
300	0.12	0.562	1.375	1340	1360	1380	1400
300	0.15	0.562	1.375	1045	1075	1105	1135
300	0.18	0.562	1.625	1341	1361	1381	1401
300	0.22	0.562	1.625	1046	1076	1106	1136
300	0.27	0.562	1.875	1342	1362	1382	1402
300	0.33	0.562	1.875	1047	1077	1107	1137
300	0.39	0.670	1.625	1343	1363	1383	1403
300	0.47	0.670	1.625	1048	1078	1108	1138
295	0.56	0.750	1.875	1344	1364	1384	1404
290	0.68	0.750	1.875	1049	1079	1109	1139
280	0.82	0.750	2.125	1345	1365	1385	1405
270	1.00	0.750	2.125	1050	1080	1110	1140
235	1.50	1.000	1.875	1051	1081	1111	1141
200	2.20	1.000	2.625	1052	1082	1112	1142
165	0.10	0.312	0.875	1031	1061	1091	1121
165	0.12	0.312	1.125	1346	1366	1386	1406
165	0.15	0.312	1.125	1032	1062	1092	1122
165	0.18	0.400	0.875	1347	1367	1387	1407
165	0.22	0.400	0.875	1033	1063	1093	1123
165	0.27	0.400	1.125	1348	1368	1388	1408
165	0.33	0.400	1.125	1034	1064	1094	1124
165	0.39	0.400	1.375	1349	1369	1389	1409
165	0.47	0.400	1.375	1035	1065	1095	1125
165	0.56	0.562	1.125	1350	1370	1390	1410
165	0.68	0.562	1.125	1036	1066	1096	1126

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

AC rated voltage (Volts (rms))	Capacitance value (µF)	Dimension, nominal 1/ (inches)		Dash number Failure rate level for (%/1,000 hr)			
		D ± 0.062	L + 0.031 - 0.051	M (1.0)	P (0.1)	R (0.01)	S (0.001)
165	0.82	0.562	1.375	1351	1371	1391	1411
165	1.00	0.562	1.375	1037	1067	1097	1127
155	1.50	0.562	1.625	1038	1068	1098	1128
150	2.20	0.670	1.625	1039	1069	1099	1129
145	2.50	0.670	1.875	1352	1372	1392	1412
140	3.30	0.750	1.875	1040	1070	1100	1130
135	4.00	0.750	2.125	1353	1373	1393	1413
130	4.70	0.750	2.375	1041	1071	1101	1131
110	6.80	1.000	1.875	1042	1072	1102	1132
100	8.00	1.000	2.125	1354	1374	1394	1414
90	9.00	1.000	2.375	1355	1375	1395	1415
80	10.00	1.000	2.625	1043	1073	1103	1133

1/ Metric equivalents, see table II.

TABLE II. Metric equivalents of decimal inches. 1/

Inches	mm	Inches	mm	Inches	mm
.031	0.79	.562	14.27	1.375	34.93
.051	1.30	.670	17.02	1.625	41.28
.062	1.57	.750	19.05	1.875	47.63
.312	7.92	.875	22.23	2.125	53.98
.400	10.16	1.000	25.40	2.375	60.33
.500	12.70	1.125	28.58	2.625	66.68

1/ Metric equivalents are given for general information only.

TABLE III. Insulation resistance.

In megohms:	
At +25°C ± 3°C (need not exceed) - - - - - - -	50,000
At +85°C +4°C, -0°C (need not exceed) - - -	10,000
In megohms x microfarads (minimum):	
At +25°C ± 3°C -	25,000
At +85°C +4°C, -0°C - - - - - - - - - - - - - - - -	5,000

Immersion: Method 104 of MIL-STD-202, condition C.

DWV:

Terminal to terminal: 140 percent of rated voltage on voltages of ≤ 300 volts; 120 percent of rated voltage on voltages of ≥ 350 volts.

Terminals to case: 900 V rms (60 Hz).

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IR:

Terminal to terminal: Not less than 33.3 percent of value specified in table III.

Terminals to case: 5,000 megohms, minimum.

Capacitance: Within ± 10 percent of initial value.

DF: Not to exceed 110 percent of initial limit.

Solderability: Method 208 of MIL-STD-202.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I. DC voltage equivalent to rms rating shall be applied.

Moisture resistance: Method 106 of MIL-STD-202.

DWV, IR, Capacitance, and DF: Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, conditions A and D.

Condition A: Applied force 5 pounds.

Condition D: 3 rotations of 360 degrees.

Low temperature and capacitance change with temperature:

Low temperature: -55°C +0°C, -3°C for 48 hours ± 4 hours with rated voltage applied.

Capacitance change with temperature:

At -55°C +0°C, -3°C: -3 percent, maximum.

At +85°C +4°C, -0°C: -3 percent maximum.

Fault count: Not applicable.

Fungus: Method 508 of MIL-STD-810, procedure I.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition C, except the time shall be 10 seconds ± 1 second.

IR: See table III.

Capacitance: Within ± 5 percent of initial value.

DF: Not more than 1.0 percent.

Life: Method 108 of MIL-STD-202.

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TABLE IV. Substitutability data.

Dash number in MIL-C-39022/7A thru MIL-PRF-39022/7D 1/					Type designation in MIL-C-39022/7 2/
L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)	
1151	1171	1191	1211	1231	
1250	1270	1290	1310	1330	
1152	1172	1192	1212	1232	
1251	1271	1291	1311	1331	
1153	1173	1193	1213	1233	
1252	1272	1292	1312	1332	
1154	1174	1194	1214	1234	
1253	1273	1293	1313	1333	
1155	1175	1195	1215	1235	
1254	1274	1294	1314	1334	
1156	1176	1196	1216	1236	
1255	1275	1295	1315	1335	
1157	1177	1197	1217	1237	
1158	1178	1198	1218	1238	
1159	1179	1199	1219	1239	
1256	1276	1296	1316	1336	
1160	1180	1200	1220	1240	
1257	1277	1297	1317	1337	
1163	1183	1203	1223	1243	
1258	1278	1298	1318	1338	
1161	1181	1201	1221	1241	
1259	1279	1299	1319	1339	
1162	1182	1202	1222	1242	
1014	1044	1074	1104	1134	CHR49A1QD104K-
	1340	1360	1380	1400	
1015	1045	1075	1105	1135	CHR49A1QD154K-
	1341	1361	1381	1401	
1016	1046	1076	1106	1136	CHR49A1QD224K-
	1342	1362	1382	1402	
1017	1047	1077	1107	1137	CHR49A1QD334K-
	1343	1363	1383	1403	
1018	1048	1078	1108	1138	CHR49A1QD474K-
	1344	1364	1384	1404	
1019	1049	1079	1109	1139	CHR49A1QZ684K-
	1345	1365	1385	1405	
1020	1050	1080	1110	1140	CHR49A1QY105K-
1021	1051	1081	1111	1141	CHR49A1QX155K-
1022	1052	1082	1112	1142	CHR49A1QC225K-
1001	1031	1061	1091	1121	CHR49A1QW104K-
	1346	1366	1386	1406	
1002	1032	1062	1092	1122	CHR49A1QW154K-
	1347	1367	1387	1407	
1003	1033	1063	1093	1123	CHR49A1QW224K-
	1348	1368	1388	1408	
1004	1034	1064	1094	1124	CHR49A1QW334K-
	1349	1369	1389	1409	
1005	1035	1065	1095	1125	CHR49A1QW474K-
	1350	1370	1390	1410	
1006	1036	1066	1096	1126	CHR49A1QW684K-
1007	1351	1371	1391	1411	

See footnotes at end of table.

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TABLE IV. Substitutability data - Continued.

Dash number in MIL-C-39022/7A thru MIL-PRF-39022/7D 1/					Type designation in MIL-C-39022/7 2/
L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)	
1008	1037	1067	1097	1127	CHR49A1QW105K-
1009	1038	1068	1098	1128	CHR49A1QU155K-
	1039	1069	1099	1129	CHR49A1QV225K-
1010	1352	1372	1392	1412	
	1040	1070	1110	1130	CHR49A1QT335K-
1011	1353	1373	1393	1413	
1012	1041	1071	1101	1131	CHR49A1QS475K-
	1042	1072	1102	1132	CHR49A1QR585K-
	1354	1374	1394	1414	
1013	1355	1375	1395	1415	
	1043	1073	1103	1133	CHR49A1QP106K-

1/ FRL L for revision C only.
2/ Complete type designation will include additional symbols for capacitance tolerance and failure rate level.

Qualification:

Accelerated conditions: 140 percent of rated ac voltage (400 Hz) for 2,000 +72, -0 hours on voltages of ≤ 300 volts; 120 percent of rated ac voltage (400 Hz) for 2,000 +72, -0 hours on voltages of ≥ 350 volts.

Rated conditions: 100 percent of rated voltage for 10,000 +96, -0 hours.

DF (at +85°C +4°C, -0°C) between 24 and 48 hours of test: Not greater than initial limit.

During last 48 hours of test: Not greater than 0.75 percent.

IR:

Terminal to terminal: Not less than 1,000 megohm/microfarads. Need not exceed 5,000 megohms.

Terminals to case: 5,000 megohms, minimum.

Capacitance: Within ± 10 percent of initial value.

DF (at +25°C ± 3°C after life): Not greater than 0.75 percent.

Quality conformance: 240 hours at accelerated conditions and meet the requirements for qualification test.

Extended life:

Accelerated conditions: 1,760 +72, -0 hours.

Marking: In accordance with MIL-PRF-39022.

Part or Identifying Number (PIN): M39022/07- (dash number from table I).

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

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